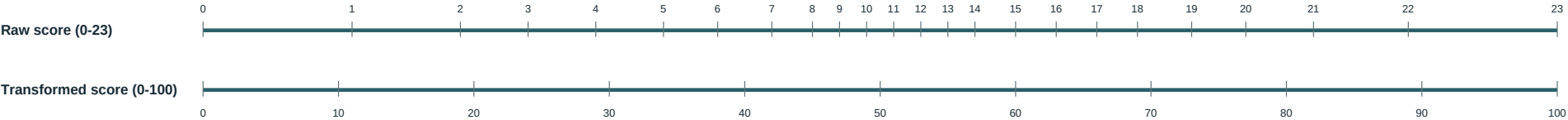


CRS-R Change Interpretation: Quick Reference

Use serial CRS-R findings, state-of-consciousness criteria, equal-interval measurement, and measurement error together when interpreting change.

Why transform the total score?

Raw CRS-R totals are ordinal: adjacent raw scores are not equally spaced. The 0-100 transformation places scores on an equal-interval ruler.



1. Start with the clinical picture

State transitions remain essential. A patient can cross a diagnostic boundary with little total-score movement, or demonstrate measurable improvement while remaining within the same state. Review the behaviors driving the score change, assessment conditions, and the broader clinical picture.

2. Quantify change on the 0-100 measure

MDC90 = 9 units | MDC95 = 11 units

These global thresholds apply one cutoff across the full transformed measure. A difference meeting or exceeding the threshold is larger than expected measurement error at the selected confidence level.

3. Use conditional change when score location matters

cMDC95 (published as cMDD95) accounts for differences in measurement precision across the measure. Measurement error is greater near the low and high extremes, so the amount of change needed to exceed error depends on the specific scores being compared.

How to use the lookup table

- Find the **initial raw CRS-R total** in the first column.
- Convert it to the corresponding **0-100 transformed measure**.
- Convert the follow-up raw score to its 0-100 measure using the same table.
- Compare the follow-up measure with the three interpretation ranges in that initial-score row: **decline beyond error**, **within measurement error**, or **improvement beyond error**.

Example: Initial raw score 2 = 19/100. A follow-up transformed measure of 0-37 is within measurement error; ≥ 38 indicates improvement beyond measurement error. No decline threshold is available from this starting score.

Detectable does not equal meaningful. Change beyond measurement error supports confidence that an observed difference reflects true change rather than expected measurement variability. It does not, by itself, establish whether that change is meaningful to the patient or care partner.

Suggested documentation: "The patient demonstrated a [X]-unit increase/decrease on the transformed 0-100 CRS-R measure, [exceeding/not exceeding] the selected measurement-error threshold."

Conditional change lookup (95% confidence)

Interpret the **follow-up transformed measure** using the row for the **initial raw score**.

Raw 0-23	Measure 0-100	cSE	Decline beyond error	Within error	Improvement beyond error
0	0	15	-	0-33	≥ 34
1	11	9	-	0-33	≥ 34
2	19	7	-	0-37	≥ 38
3	24	6	-	0-41	≥ 42
4	29	6	-	0-44	≥ 45
5	34	6	≤ 18	19-48	≥ 49
6	38	6	≤ 23	24-52	≥ 53
7	42	5	≤ 28	29-54	≥ 55
8	45	5	≤ 33	34-56	≥ 57
9	47	4	≤ 33	34-59	≥ 60
10	49	4	≤ 37	38-62	≥ 63
11	51	4	≤ 37	38-65	≥ 66
12	53	4	≤ 41	42-65	≥ 66
13	55	4	≤ 44	45-68	≥ 69
14	57	4	≤ 46	47-72	≥ 73
15	60	5	≤ 48	49-76	≥ 77
16	63	5	≤ 50	51-81	≥ 82
17	66	5	≤ 54	55-88	≥ 89
18	69	5	≤ 56	57-88	≥ 89
19	73	6	≤ 59	60-100	-
20	77	6	≤ 62	63-100	-
21	82	7	≤ 65	66-100	-
22	89	9	≤ 72	73-100	-
23	100	15	≤ 72	73-100	-

cSE = conditional standard error. Ranges and transformed values reproduced from Table 4 of the supplied Better Measurement manuscript/paper.

Sources: Weaver et al., *Journal of Neurotrauma* (2024), Interpreting Change in Disorders of Consciousness Using the CRS-R; Grady-Dominguez et al., *Archives of Physical Medicine and Rehabilitation* (Better Measurement special communication). **Research/team-review prototype - not a substitute for clinical judgment.**